

Stone–Jordan Exceptional Theory

Building the Universe from the Mirror

Brandon Belna

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Abstract

We construct a mathematical model of the universe from two primitives: the **void** (no observable distinction) and an involutive **mirror operator** M on profinite observable spectra. Iterating M climbs the Hurwitz division ladder and the exceptional Lie cascade; capacity balancing $\text{Bal}_{\mathcal{L}}$ is the dual mirror selecting unique partitions at each rung. The universe is the profinite fixed point $\mathfrak{U}^* = C^\infty(\mathcal{V})$, stabilizing on the Albert algebra $J_3(\mathbb{O})$ and the compact exceptional group E_6 . Physical observables are σ -anti-invariant modes on the mirror quotient at the heterotic rung; soldering, dynamics, and gauge data follow from the Jordan–octonionic block $H_2(\mathbb{O})$. We state proved results (mirror rung table, three-family multiplicity, anomaly cancellation, induced gravitation) and isolate conjectural steps (full GUT breaking, nonperturbative reflection positivity, cosmological constant) without overclaiming a completed physical theory.

Keywords: void; mirror operator; Stone duality; exceptional Jordan algebra; profinite limit; E_6 coset.

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1 Introduction

This paper develops *Stone–Jordan Exceptional Theory* (SJET) as a **mathematical physics** construction: starting from a void and a mirror, we build a universe datum $\mathfrak{U}^*$ with Lie–Jordan structure, capacity-selected dynamics, and a contact layer with four-dimensional gauge theory. The mirror is not metaphor; it is a functorial involution on Stone spectra whose orbit generates all arithmetic and exceptional data.

1.1 Contributions

- (i) **Two axioms only:** void $\mathcal{V}$ and mirror M .
- (ii) **Universe construction:** $\mathfrak{U}^* = \varprojlim_n (\text{Bal}_{\mathcal{C}} \circ M)^n(\mathcal{V})$.
- (iii) **Proved mirror rungs** through E_6 and chart synchronization on exceptional rungs E_8 , $E_8 \times E_8$.
- (iv) **Physical sector** as mirror-quotient σ -anti-invariant cohomology; three families without a separate family postulate.
- (v) **Honest epistemic scope:** theorems, derived definitions, and named conjectures for remaining physical inputs.

1.2 Conventions

Compact real E_6 ; order parameter $\Phi \in J_3(\mathbb{O})$; capacity functional $\mathcal{C}$. Branching $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$ under $H = \text{Spin}(10) \times U(1)$. Signature $(-, +, +, +)$ when a Lorentzian soldering map is fixed (Section 5.2).

1.3 Organization

Section 2: axioms and mirror functor. Section 3: division and exceptional ladders. Section 4: capacity, potential, universe fixed point. Section 5: physical sector, spacetime, gauge, matter. Section 6: status and open problems. Appendix A: Albert identities.

2 The Void and the Mirror

2.1 The void

Definition 2.1 (Void). The *void* $\mathcal{V}$ is the trivial Boolean algebra $\mathcal{B}_0 = \{\perp, \top\}$ with Stone spectrum a single point. The associated algebra is $\{0\}$; no local quantum number exists.

Axiom 2.2 (Void). Physical structure arises by departure from $\mathcal{V}$. No graph, Lie group, or space-time manifold is primitive.

2.2 The mirror operator

Definition 2.3 (Mirror modes). At rung n , M acts in one of four modes:

M1. Boolean doubling ($0 \rightarrow 1$): adjoin one generator; $|S_{n+1}| = 2|S_n|$.

M2. Cayley–Dickson ($1 \rightarrow 4$): $D_{n+1} = D_n \oplus D_n$; dimensions 1, 2, 4, 8.

M3. Jordan exit ($4 \rightarrow 5$): sedenions fail; exit to $J_3(\mathbb{O}) = H_3(\mathbb{O})$, $d_5 = 27$.

M4. Exceptional (≥ 6): automorphism, product, or Cartan mirrors on Lie data (Section 3.2).

Definition 2.4 (Mirror operator). A *mirror operator* is a functor $\mathbf{M}$ on a category of Stone rungs such that: (i) $\mathbf{M}^2 \cong \text{id}$ on stabilized rungs; (ii) $\mathbf{M}(\mathcal{V}) \neq \mathcal{V}$; (iii) $\mathbf{M}$ is functorial on morphisms.

Axiom 2.5 (Mirror). There exists $\mathbf{M}$ as in Definition 2.4 whose orbit from $\mathcal{V}$ generates the unified climb (3.1). At each rung, $\text{Bal}_{\mathcal{C}}$ balances mirror-refined partitions (Definition 4.1).

Definition 2.6 (Climb and mirror power). $\mathbf{C} = \text{Bal}_{\mathcal{C}} \circ \mathbf{M}$ and $(S_n, A_n) = \mathbf{M}^n(S_0, A_0)$ with $(S_0, A_0) = (S(\mathcal{V}), \{0\})$.

Proposition 2.7 (Involution). *On stabilized rungs, $\mathbf{M}^2 \cong \text{id}$ up to gauge: Boolean complement, field conjugation, product factor exchange, and coset mirror $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$.*

3 The Mirror Climb

$$\mathcal{V} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow J_3(\mathbb{O}) \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \cdots \rightarrow \mathcal{V} \quad (3.1)$$

3.1 Division rungs

n	A_n	d_n	Mode
0	$\{0\}$	0	<i>void</i>
1	$\mathbb{R}$	1	<i>discrete</i>
2	$\mathbb{C}$	2	<i>arithmetic</i>
3	$\mathbb{H}$	4	<i>arithmetic</i>
4	$\mathbb{O}$	8	<i>Hurwitz terminal</i>
5	$J_3(\mathbb{O})$	27	<i>Jordan exit</i>

Theorem 3.1 (Mirror rung table, $n \leq 5$).

In particular $\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$ and $\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$.

Proof. Cayley–Dickson doubling until $\mathbb{O}$; Hurwitz uniqueness of normed division algebras; sedenion zero divisors force Jordan exit to the unique formally real $H_3(\mathbb{O})$ [2, 5]. $\square$

Lemma 3.2 (Sedenion failure). $\mathbf{M}_{\text{arith}}(\mathbb{O})$ is 16-dimensional and not division.

3.2 Exceptional rungs

On $J_3(\mathbb{O})$, the stabilizer tower is $G_2 \subset F_4 \subset E_6$. The Freudenthal diagonal $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$ yields:

Theorem 3.3 ($\mathbf{M}^6(\mathcal{V}) = E_8$). *Automorphism mirror on the octonion–Jordan pair gives E_8 , $d_6 = 248$.*

Theorem 3.4 ($\mathbf{M}^7(\mathcal{V}) = E_8 \times E_8$). *Product mirror $\mathbf{M}_{\text{prod}}(G) = G \times G$ gives $d_7 = 496$.*

Theorem 3.5 ($\mathbf{M}^8(\mathcal{V}) = E_6$). *Cartan mirror after capacity descent on the Albert graph gives $d_8 = 78$, $H = \text{Spin}(10) \times U(1)$, $\mathcal{M} = E_6/H$ (EIII), $\dim \mathcal{M} = 32$.*

Proof sketches. E_8 : couple pre-projection octonion data to $J_3(\mathbb{O})$ via Freudenthal–Tits [3, 4]. $E_8 \times E_8$: functorial product involution. E_6 : capacity-balanced collapse of heterotic product to reduced structure group of $J_3(\mathbb{O})$ with branching $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$. $\square$

Definition 3.6 (Chart dimension). At $n \geq 6$, $\chi(S_n) = \dim A_n \in \{248, 496, 78\}$; profinite $|S_n|$ is generally infinite. Stone refinement is chart synchronization, not cardinality matching [1].

4 Building the Universe

The universe is not assumed. It is the profinite fixed point of mirror climb with capacity balancing.

4.1 Capacity dual

Definition 4.1 (Albert graph and capacity). On the mirror-refined Albert–Cayley graph $\mathcal{G}$ with costs c_i ,

$$\mathcal{C}(\omega) = - \sum_{a \in V} \mu(a) \log \left(\sum_i e^{c_i(a) - \omega_i} \right). \quad (4.1)$$

$\text{Bal}_{\mathcal{C}}$ is the unique maximizer on $\{\sum_i \omega_i = 0\}$.

Theorem 4.2 (Concavity). $\mathcal{C}$ is strictly concave; for targets $m_i > 0$, $\sum m_i = 1$, there is unique ω^* with $\partial \mathcal{C} / \partial \omega_i(\omega^*) = m_i$.

Theorem 4.3 (1|10|16 partition). At $M^5(\mathcal{V})$, $\text{Bal}_{\mathcal{C}}$ yields sector masses $(1/27, 10/27, 16/27)$ on $27 = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$.

4.2 Potential and vacuum

Definition 4.4 (Potential).

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \text{ even}, \quad (4.2)$$

is the profinite limit of graph penalties under $\text{Bal}_{\mathcal{C}}$.

Theorem 4.5 (EIII vacuum). V is coercive on compact E_6 . Global minima lie on rank-one EIII $\mathcal{M} = E_6/H$; the origin is a saddle with 32 Goldstone directions.

4.3 Universe datum

Definition 4.6 (Universe). The *universe datum* is

$$\mathfrak{U}^* = C^\infty(\mathcal{V}) = \varprojlim_{n \rightarrow \infty} C^n(\mathcal{V}), \quad (4.3)$$

stabilizing at $J_3(\mathbb{O})$ with Albert graph $\mathcal{G}_{\text{Albert}}$, capacity maximizer ω^* , and vacuum $\Phi_\star \in \mathcal{M}$.

Theorem 4.7 (Universe from the mirror). $\mathfrak{U}^*$ carries: (i) 27 stable Jordan coordinates; (ii) 1|10|16 capacity partition; (iii) E_6 gauge sector with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$; (iv) EIII coset $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$.

Proof. Items (i)–(ii) from Theorems 3.1 and 4.3. Items (iii)–(iv) from Theorem 3.5 and standard E_6 branching. $\square$

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[\frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \quad (4.4)$$

with K the unique H -invariant metric on $\mathfrak{m}$.

5 The Physical Universe

Mathematical data of $\mathfrak{U}^*$ becomes physical through mirror quotient observability and soldering on the Jordan–octonionic block.

5.1 Physical sector

At $M^7(\mathcal{V}) = E_8 \times E_8$, let Q be the Stone lift of the stabilized Albert tower and σ the product-mirror involution $(g_1, g_2) \mapsto (g_2, g_1)$.

Definition 5.1 (Physical configurations). A configuration is *observable* if it lies in the σ -anti-invariant sector of $\hat{Q} = Q/\sigma$. Fermions are cohomology classes in $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ for the **16**-bundle V_{16} .

Theorem 5.2 (Three families). $\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3$. Each class is one chiral **16** of $\text{Spin}(10)$; mirror partners are σ -invariant and non-observable. Hence $n_F = 3$.

Proof. Cellular count on the rank-3 diagonal $\mathfrak{d} \subset J_3(\mathbb{O})$; index theorem on $\hat{Q}$ [7]. See Appendix A.2. $\square$

5.2 Spacetime from the climb

Definition 5.3 (Soldering). Embed $\mathbb{H} \subset \mathbb{O}$ from the mirror-fixed associative subalgebra at the Jordan exit. The block $H_2(\mathbb{H}) \subset H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ carries Lorentzian signature via $\det$. The soldering map σ identifies four clock components $\Phi^0, \dots, \Phi^3$ with this block; $d = 4$.

Proposition 5.4 (Coset metric). $K|_{\mathfrak{m}}$ is positive-definite and H -invariant. Lorentzian signature is carried by σ , not by Wick rotation of K .

Theorem 5.5 (Semiclassical reflection positivity). On *EIII*, with radiative mass $\mu^2 > 0$ on 28 coset scalars (Coleman–Weinberg), the Euclidean σ -model (4.4) satisfies reflection positivity along Φ^0 in the semiclassical regime.

Conjecture 5.6 (Nonperturbative reflection positivity). Tree-level Goldstones and the nonperturbative gauge measure satisfy full reflection positivity, yielding OS reconstruction [9].

5.3 Gauge, anomalies, gravity

Theorem 5.7 (Anomaly cancellation). For one embedded **27**: cubic $16 - 80 + 64 = 0$; mixed coefficient -1 cancelled by Green–Schwarz [8]. With $n_F = 3$ replicas of the full **27** arithmetic, anomalies cancel family-wise.

Theorem 5.8 ($b_0 = 12$). With $C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $n_F = 3$, $n_S = 28$: $b_0 = (88 - 24 - 28)/3 = 12$.

Theorem 5.9 (Induced Planck mass). Sakharov induction from 28 massive coset scalars gives $M_{\text{Pl}}^2 \propto 28\mu^2/(6 \cdot 16\pi^2) \log(\Lambda^2/\mu^2)$.

Theorem 5.10 (Gravitational fixed point). One-loop truncation gives $\kappa_\star = 96\pi^2/5$ with $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$.

Conjecture 5.11 (GUT breaking). Gradient flow of V on coset directions triggers $E_6 \rightarrow \text{Spin}(10) \times U(1) \rightarrow \dots \rightarrow SU(3)_c \times U(1)_{\text{em}}$ with Higgs content **45_H**, **126_H**, **10_H**.

6 Conclusions

SJET builds the universe from the mirror:

$$\mathcal{V} \xrightarrow{M} \dots \xrightarrow{M} \mathfrak{U}^\star, \quad \mathfrak{U}^\star = C^\infty(\mathcal{V}). \quad (6.1)$$

Only two axioms are primitive. Capacity balancing, the E_6 potential, gauge structure, family multiplicity, and semiclassical unitarity are **derived** or **defined** from mirror data.

6.1 Epistemic summary

Status	Content
Axioms	void, mirror
Theorems	rung table, E_8 , $E_8 \times E_8$, E_6 , partition, vacuum, three families, anomalies, $b_0 = 12$, κ_* , semiclassical RP
Definitions	physical sector, soldering, universe $\mathfrak{U}^*$
Conjectures	full RP, GUT breaking, CC sequestering

6.2 Open problems

- O1.** Functorial uniqueness of soldering (Definition 5.3).
- O2.** Prove Conjecture 5.11 from V .
- O3.** Prove Conjecture 5.6.
- O4.** Graviton/ R^2 persistence of κ_* .
- O5.** Cosmological constant from σ -invariant sequestering at κ_* .

6.3 Archive

The prior modular manuscript (39 pages, verification ledger, axiomatic elimination program) is preserved in `old/sjet-stack/`.

A Albert algebra and mirror quotient

A.1 Albert model

Elements of $J_3(\mathbb{O})$ are 3×3 Hermitian octonionic matrices

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbb{R}, \quad x_i \in \mathbb{O}. \quad (\text{A.1})$$

The sharp product $X^\#$ and cubic norm $N(X) = \det X$ define the E_6 structure group [5, 6].

A.2 Mirror quotient cohomology

On the stabilized tower $\mathcal{T}_7$ at $M^7(\mathcal{V})$, cellular 1-cochains on three diagonal cells give $\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3$. The Stone lift identifies this with $H^1(\hat{Q}, V_{16})^{\sigma=-1}$, proving Theorem 5.2.

References

- [1] M. H. Stone, The theory of representations for Boolean algebras, *Trans. Amer. Math. Soc.* **40** (1936), 37–111.
- [2] A. Hurwitz, Über die Komposition der quadratischen Formen, *Math. Ann.* **88** (1923), 1–25.
- [3] H. Freudenthal, Beziehungen der $\mathfrak{E}_7$ und $\mathfrak{E}_8$ zur Oktavenebene, *Indag. Math.* **16** (1954), 218–230.

- [4] J. Tits, Algèbres alternatives, algèbres de Jordan et algèbres de Lie exceptionnelles, *Indag. Math.* **28** (1966), 223–237.
- [5] T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer, 2000.
- [6] N. Jacobson, *Structure and Representations of Jordan Algebras*, AMS, 1968.
- [7] M. F. Atiyah and R. Bott, A Lefschetz fixed point formula for elliptic complexes I, *Ann. of Math.* **86** (1967), 374–407.
- [8] M. B. Green and J. H. Schwarz, Anomaly cancellations in supersymmetric $D = 10$ gauge theory, *Phys. Lett. B* **149** (1984), 117–122.
- [9] K. Osterwalder and R. Schrader, Axioms for Euclidean Green’s functions, *Commun. Math. Phys.* **31** (1973), 83–112.
- [10] S. Coleman and E. Weinberg, Radiative corrections as the origin of spontaneous symmetry breaking, *Phys. Rev. D* **7** (1973), 1888–1910.
- [11] A. D. Sakharov, Vacuum quantum fluctuations in curved space and the theory of gravitation, *Dokl. Akad. Nauk SSSR* **177** (1967), 70–71.
- [12] E. Witten, Symmetry breaking patterns in superstring models, *Nucl. Phys. B* **258** (1985), 75–100.